OpenGEN-1: A Reconstruction of GEN-1 from First Principles

This is a reconstruction of the GEN-1 model from first principles, supported by evidence posted by Generalist AI, demos, blogs, and Pete / Andy's blogs and interviews.

The core claim is:

GEN-1 is a physical foundation model trained mainly on streams of physical interaction, not a VLA trained from a VLM.

Pete Florence and the Generalist team make this distinction explicit: GEN-1 is "not a fine-tuned vision-language model with robot actions bolted on," and is also "not just a world model" [1]. They instead call it a native foundation model for physical interaction [1].

That phrase matters. It implies the model is not mainly learning language, object semantics, or passive video prediction. It is learning the structure of physical interaction: sensing, acting, contact, failure, recovery, speed, and consequence.

1. From Semantic Commonsense to Physical Commonsense

LLMs and VLMs trained on internet-scale data can absorb tremendous semantic knowledge. They know what an apple is, what color it usually is, where it is found, and what people usually do with it.

But they are missing something deeper: **physical commonsense** [2].

If I push an apple left, it may roll left. If there is a box in front of it, it will be blocked. If I hold it and it slips, I tighten my grip. These are not merely visual facts. They are learned from action and consequence.

Andy Zeng's post makes this point directly: if large-scale text gives semantic commonsense, then large-scale physical interaction may give physical commonsense — but only if the data preserves the loop [2].

There are two levels here:

- **Forward physical commonsense:** if I take this action, what will happen next?
- **Inverse / reflexive physical commonsense:** given what is happening now, what should I do?

The first is close to a dynamics model or world model. The second is closer to a policy, inverse dynamics model, or reflex. Humans do both continuously. We do not first build a perfect world model, then plan, then act. We sense and act in one loop.

So the central question is:

What model architecture and data distribution would allow physical commonsense to emerge?

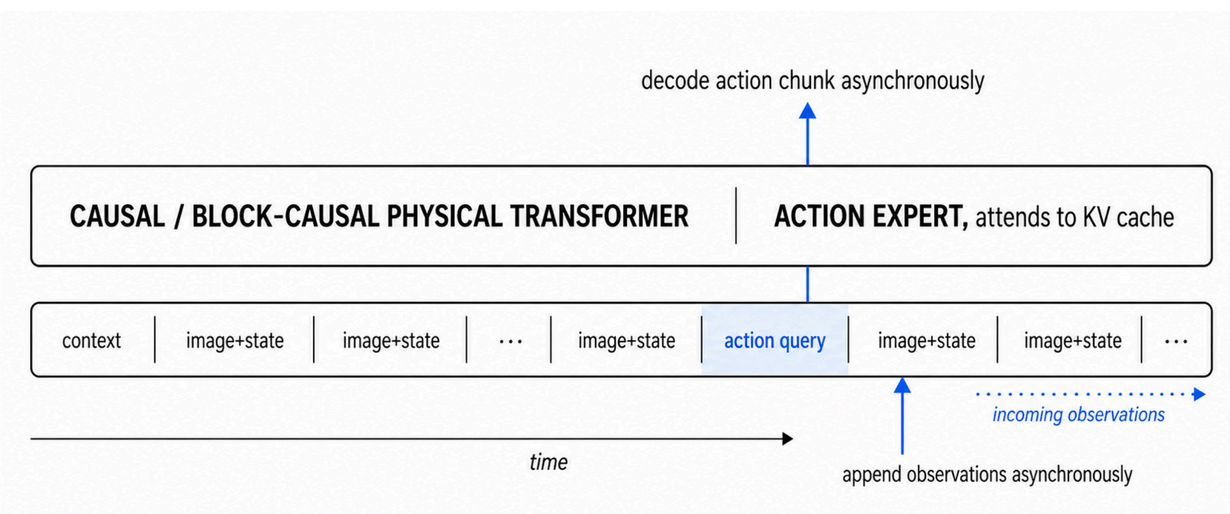
2. A Streaming Physical-Token Architecture

When interacting with the physical world, the robot receives a stream of sensory inputs and emits a stream of actions. The natural architecture is therefore not a static image-to-action network, but a streaming sequence model.

A plausible architecture is a causal or block-causal transformer over physical tokens, where each token can be:

- context / task tokens;
- image tokens;
- proprioception tokens;
- force / tactile tokens;
- action-condition tokens;
- state-query or action-query tokens.

During inference, the model continuously appends observation chunks and periodically decodes action chunks.



This is highly aligned with Generalist's own language around GEN-0. They describe Harmonic Reasoning as training models to "simultaneously think and act," through an interplay between asynchronous streams of sensing and acting tokens [3].

The KV cache of the past processed stream acts as short-term memory. A longer memory may also be provided by harness-managed keyframes or compressed state summaries. Generalist explicitly says GEN-1 required "new forms of paged attention" for real-time inference [5]. In transformer serving, PagedAttention is specifically a KV-cache memory-management method [6].

This also fits the shell-game demo. That task requires visual memory: with only wrist cameras, objects can leave the field of view, become occluded, or move behind the gripper. A current-frame policy is not enough. The model needs a short-term working memory over recent observations and states [4].

Action decoding can be implemented in several ways. The simplest is a direct action-chunk head similar to the StarVLA alpha [17]. A stronger option is an OpenPI style flow action expert, where future actions are represented as continuous noisy action tokens and the model predicts a vector field from noise to clean actions [7]. Another plausible option is a GR00T-like diffusion transformer action module, where a fast action model generates fluid motor actions conditioned on visual-language-proprioceptive context [8].

3. Training Objective: Action Prediction + Latent Dynamics

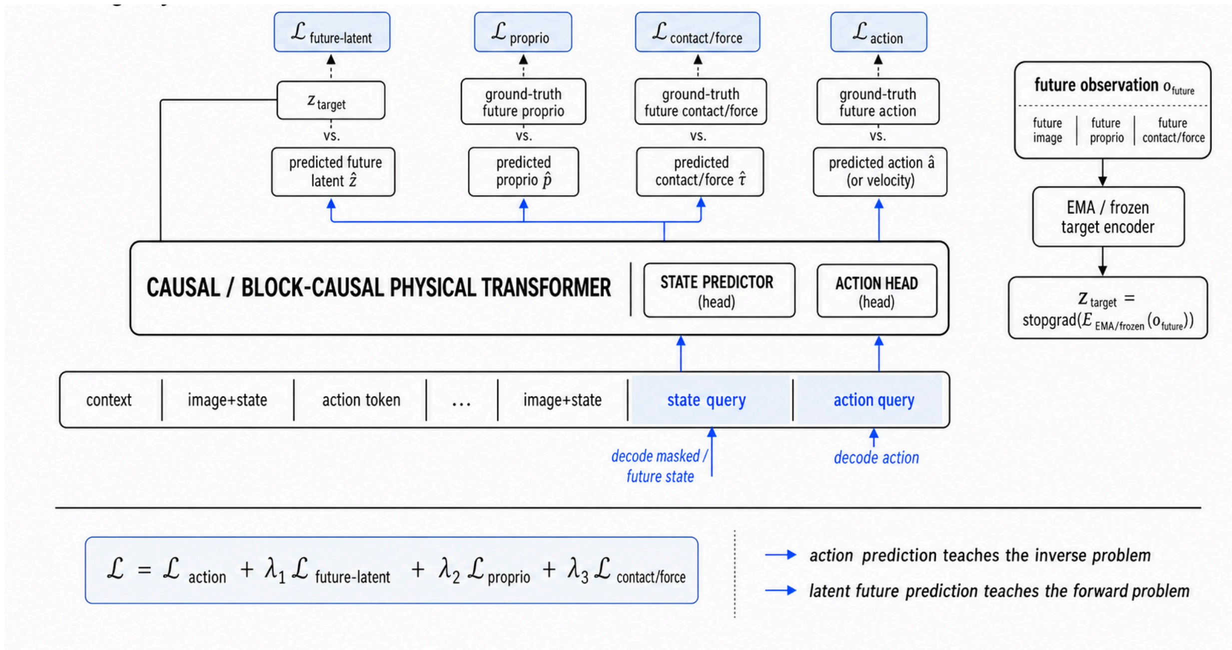
The standard VLA objective is action prediction loss that between the prediction action and ground truth actions (or velocity for the flow matching).

This teaches the inverse problem: **Given the current history, what should I do next?**

But to learn physical commonsense, action prediction alone is probably not enough. The model should also learn the forward problem: **Given the current history and an action, what physical state follows?**

This is where world-model ideas become useful. But full RGB reconstruction is likely too expensive and too distracted by irrelevant pixel details. The model does not need to reconstruct every background texture. It needs to predict action-relevant physical state.

A better objective is latent future prediction that aligns the predicted future latent state with the encoded future state.



This is the technical meaning of “VLA + world model + beyond.” Pete says Generalist has spent over a year combining ideas from “VLAs, world models, and beyond,” because the more capabilities a model combines, the harder it is to categorize [1].

For policy inference, past predicted actions are not strictly required if the model has rich image/proprio history. The state trajectory already contains much of their effect. But for world-model training, clean action-condition tokens are useful because they teach causal dynamics.

However, latent reconstruction can collapse if both the encoder and predictor are trained freely. The target latent should be anchored:

$$ z_{\text{target}}
\text{stopgrad} \left(E_{\text{EMA or frozen}}(o_{\text{future}}) \right) $$

V-JEPA-style methods use latent-space prediction, stop-gradient target encoders, masking, and predictor asymmetry to avoid collapse [9]. For a physical foundation model, the action loss itself is also an anti-collapse anchor, so the formulation can be simpler.

A concise objective could be:

$$ \mathcal{L} = \mathcal{L}_{action} + \lambda_1 \mathcal{L}_{future-latent} + \lambda_2 \mathcal{L}_{proprio} + \lambda_3 \mathcal{L}_{contact/force} $$

This objective combines inverse dynamics, latent forward dynamics, proprioceptive prediction, and contact/force prediction.

4. Why the Data Has to Be Physical Interaction Data

Internet text and video can provide semantic priors, but they do not contain the full physical loop. Most internet video lacks accurate actions, proprioception, force, grip, tactile feedback, and recovery traces.

Generalist's claim is that the data bottleneck can be broken by collecting physical interaction at scale. GEN-1 is trained on more than half a million hours of high-fidelity physical interaction data [5]. More importantly, the base foundation model is trained without robot data; instead, it uses data from low-cost wearable devices on humans doing millions of activities [5].

This matches Andy's data argument. Teleoperation often breaks the sensorimotor loop through latency, limited tactile feedback, and unnatural interfaces; Generalist instead built handheld ergonomic devices where the force feedback is present and operators start reacting rather than thinking [2].

The diversity should not only be semantic diversity — different objects, scenes, and tasks. It should include interaction diversity:

- too much force → object deforms or crashes;
- too little force → object slips;
- bad grasp → regrasp;

This is where tactile and force data become crucial. Vision tells the robot where things are. Proprioception tells it where the robot is. Tactile and force tell it what physical interaction is actually happening.

Generalist's blog claims GEN-1 is trained with more than 500,000 hours of interaction data [5].

Assume the data is recorded at 10 Hz: 500,000 x 3600 x 10 = 18B

So if each timestep is compressed into roughly 64–256 visual tokens, then the total tokens counts towards 1.15~4.61T. This is before adding proprioception, action, tactile, force, and task tokens. So the dataset is already in the trillion-token regime.

Compare it with common LLMs / VLMs:

Model family	**Model size**	**Pretraining tokens**	**Notes**
Llama 2	7B / 13B / 70B	2.0T text tokens	Same token count reported for all three Llama 2 sizes [10].

Model family	Model size	Pretraining tokens	Notes
Llama 3	8B / 70B	15T+ text tokens	Meta reports 15T+ pretraining tokens for both Llama 3 sizes [11].
Qwen2-VL	2B / 8B / 72B	~1.4T multimodal tokens	Open-Qwen2VL reports that Qwen2-VL used roughly 1.4T multimodal pretraining tokens [12, 13].

The dataset scale is large enough to train a 10B-class physical foundation model, especially because the tokens are dense in causal physical information.

That aligns with GEN-0's scaling observations: Generalist reports a phase transition around 7B parameters, says 7B+ models internalize large-scale robotic pretraining data better, and notes that GEN-0 has been scaled to 10B+ model sizes [3].

The key point is not simply token count. A trillion web tokens and a trillion physical-interaction tokens are not equivalent. Web tokens are dense in semantic information. Physical-interaction tokens are dense in contact, action, failure, correction, memory, and consequence.

6. Inference at 100 Hz Requires a System

Generalist's earlier research preview says the robot maps pixels and other sensor data to 100 Hz actions, and that the full hardware/software stack enables reactive, smooth, precise control [16].

For a 10B model in fp16/bf16, the weights alone are about 20GB. At batch size 1, next-token or action-query decoding is often memory-bandwidth-bound. Ignoring overhead.

Approximate bandwidth-only estimates:

GPU	Memory bandwidth	fp16 10B weight-read lower bound	More realistic optimized range
RTX 5090	~1.79 TB/s	~11 ms	~10–25 ms
RTX PRO 6000 Blackwell	~1.79 TB/s	~11 ms	~10–25 ms
L40S	864 GB/s	~23 ms	~25–60 ms

These are only weight-read lower bounds. Real inference also pays for KV-cache reads, vision encoding, action expert steps, kernel overhead, scheduling, robot I/O, and safety/control logic.

So the likely design is:

Large model:

- runs at lower policy frequency;
- maintains memory through KV cache / paged attention;
- decodes action chunks;
- predicts or tracks latent physical state.

Low-level harness:

- executes at 100 Hz;
- blends action chunks;
- enforces safety;
- handles impedance / force / gripper control;
- reacts to tactile and proprioceptive events;
- smooths and stabilizes behavior between model calls.

This explains why Generalist says GEN-1 is more accurately a system than just a model [5].

7. Conclusion

The key hidden insight is:

GEN-1 is not just a better robot policy. It is an attempt to build a pretraining substrate for physical interaction.

This is in contrast to other VLAs that initialize from pretrained VLMs and use large amounts of web data to co-train the model, which teaches semantic commonsense. GEN-1 appears to target the missing layer: physical commonsense learned from dense streams of real interaction.

Demos from Generalist mostly focus on showcasing generalization over physical commonsense: contact-rich manipulation, occlusion, recovery, smoothness. By contrast, many π-series / VLM-based robot models emphasize semantic or task-level generalization: following instructions, unseen objects, and transferring across high-level task descriptions.

References

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